GLOBAL INSTITUTE OF TECHNOLOGY

B. Tech. I Semester 2024-25

1FY3-07: Basic Mechanical Engineering

Branch-All

Guess Paper

Part A(Answer should be given up to 25 words only)

All questions are compulsory

Q.1 State Zeroth Law and First Law of Thermodynamics.

Q.2 What is forging?

Q.3 Name different types of power plant.

Q.4 Distinguish between a heat engine and a refrigerator.

Q.5 Write different types of gears.

Q.6 Write the difference between Brazing and Soldering.

Q.7 Classify the IC engines.

Q.8 What is the application of boiler in industry?

Q.9 Write difference between manufacturing engineering and design engineering.

Q.10 Define any two 'mechanical properties' of materials.

Q.11 Describe with figure different type of belt drive.

Q.12 Define the coefficient of performance of Refrigerator

Q.13 What is the pattern in casting process?

Q.14 What is industrial engineering & its scope?

Q.15.Differentiate between water tube and fire tube boiler?

Q.16 Write a short note on different type of power plants.

Q.17 What is the IP & BP in the Internal Combustion Engine?

Q.18. Describe modern tools used in Mechanical Engineering.

Q.19 Differentiate between impulse and reaction turbine?

Q.20 State the first law of thermodynamics.

Q 21. What do you mean by slip in belt?

Q22.Which pump require priming? What is the need for priming.

Q23. Differentiate between brazing and soldering.

Q24. Mention the steps involved in designing of a part (Product)

Q25. Differentiate between COP and efficiency of system.

Q 26. Write two important properties of steam.

Q 27. List the four important properties of moulding sand.

Q 28. Give the example of any two allays and state their application.

Q29. Mention the different fields of mechanical engineering.

Q 30. Write name of different types of power transmission devices.

Part B Analytical/Problem solving questions

Attempt all questions (word Limit 75)

Q.1 Differentiate between two-stroke and four-stroke I.C. engines.?
Q.2 Explain the gas welding process,
Q.3 Write short notes on ductility, annealing and toughness.
Q.4 Explain the working of 4 stroke S1 engine (petrol engine). Also write all component of SI engine (with neat and clean diagram).
Q.5 Write classification and types of refrigeration systems and air- conditioning. Also explain the working of domestic refrigerator (with neat and clean diagram)
Q.6 Explain the Metal Casting Process in detail. Also explain any 5 tools used in the casting process.
0.7 Derive the expression for length of the cross-belt transmission.
Q.8 Classify steam boilers. Explain the construction details and working of Cochran boiler with neat sketch.
Q.9. Explain any one type of water tube boiler with neat sketch.
Q.10 Explain differentiate between 2 stroke & 4 stroke engines.
Q.11 How Cavitation can be eliminated by Pump?
Q.12 Describe with figure different types of belt drive?
Q.13. What is-air conditioning? Draw and describe different component used in it
Q.14. Explain the various stages of Heat treatment process?
Q.15 What is compounding in steam turbine? Explain various types of compounding in impulse steam turbine with suitable schematic diagrams.
Q.16. What are the important differences between reciprocating pump and centrifugal pump? Also gives their applications.
Q.17 Compare and contrast the working of vapour compression and vapour absorption refrigeration systems. Draw schematic diagram of each.
Q.18.Describe the different types of gears with sketches.
Q19. A Carnot refrigerator system has a working temperature of -30°C and 40°C. What is the maximum C.O.P. possible? If the actual C.O.P. is 75% of maximum, calculate the actual refrigeration effect produced per kWh and the capacity of the system.

Part C(Descriptive/Analytical/Problem Solving/Design Question)

Attempt all questions(word Limit 125)

Q.1 Discuss the various metal forming processes in detail.

Q.2 An engine running at 150 r.p.m. drives a line shaft by means of a belt. The engine pulley is 750 mm in diameter and the pulley on the line shaft is 450 mm. A 900 mm diameter pulley on the line shaft drives a 150 mm diameter pulley keyed to a dynanio shaft. Find the speed of the dynamo shaft when;

1. there is no slip, and

2. there is a slip of 2% at each drive.

Q.3.Explain the working of 4 stroke S1 engine (petrol engine). Also write all component of SI engine (with neat and clean diagram).

Q.4 Write classification and types of refrigeration systems and air- conditioning. Also explain the working of domestic refrigerator (with neat and clean diagram)

Q.5.Explain the Metal Casting Process in detail. Also explain any 5 tools used in the casting process.

0.6.Derive the expression for length of the cross-belt transmission.

Q.7 Classify steam boilers. Explain the construction details and working of Cochran boiler with neat sketch.

Q.8 What is meant by refrigeration system? Describe vapor compression refrigeration system?

Q.9 What is gear transmission? Describe different types of gear.

Q.10 Describe rolling process with neat sketches.

Q.11 With a suitable sketch explain the working of centrifugal pump.

Q.12 Describe hardening and tempering of steel.

Q.13 What is Nuclear power plant? Explain the various components of Nuclear power plant.Describe the working of Nuclear power plant, advantages and disadvantages,

Q.14 An engine, based on air standard otto cycle, is supplied with air at 0.1 MPa and 35°C.The compression ratio is 8. The heat supplied is 500 kJ/kg. For given working air specific heat capacity at constant pressure and at constant volume is 1.005 kJ/kgK 0.718kJ/kgK respectively, find the following-

(a) Efficiency of an engine

(b) Temperature and pressure at the end of tompression

(c) Maximum temperature of the dyed

Q.15 What is Open Belt Drive? Derive the expression for the length of belt dive .